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Review article

The role of digital health in supporting cancer patients' mental health and psychological well-being for a better quality of life: A systematic literature review

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ABSTRACT

Background: This work aims to evaluate the role of digital health in supporting the mental and psychological well-being of patients with cancer and identify the associated challenges of use and implementation.
Methods: Eligibility criteria: We included peer-reviewed studies (quantitative/qualitative) published between January 2011 and July 2022, that are written in English using technology to support cancer patients' mental health. We excluded opinion papers, editorials, and commentaries. *Information sources:* The systematic review was conducted across ProQuest CENTRAL, Scopus, PubMed, PsycInfo, Web Of Science, and IEEE Xplore. The study selection followed the Preferred Reporting Items for Systematic Reviews, meta-Analysis Reviews, and meta-Analysis guidelines (PRISMA). *Risk of bias:* All screening steps followed a consensus between the authors to minimize bias or discrepancy. *Synthesis of the results:* Data were extracted following the Six-factor Model of Psychological Well-being (SMPW). The technology challenges are summarized following the Systems Engineering Initiative for Patient Safety model (SEIPS), focusing on design, impact on processes, and outcomes.
Results: We included 25 studies satisfying our inclusion criteria. The studies had little interest in minorities and sociodemographic factors' assessment within their results. The review showed that mental health and psychological well-being tools cover many applications. In addition to allowing personal growth, digital health can help cancer patients gain more autonomy and self-acceptance. Moreover, these health technologies can aid in mastering the environment, shaping social relationships, and pursuing life goals. Many challenges were identified related to the environment, organization, users, and tasks.
Discussion: Digital health applications for cancer care cover a broad spectrum of mental health interventions. Challenges warrant analyzing the needs and usability. Lessons learned during COVID-19 may help refine technology interventions for mental health in cancer care. More interest in minorities is needed when designing technologies for patients to ensure more access to equitable care.

1. Introduction

Cancer is a disease of global public health relevance, encompassing 19.3 million new cancer cases and nearly 10.0 million cancer deaths in 2020. In the United States, in 2022, 1,918,030 new cancer cases and 609,360 cancer deaths were projected [1]. For every 100,000 people, 439 new cancer cases were reported, and 146 died of cancer [2]. Cancer is a multifaceted global disease that spans the breadth of human experience [3], affecting daily life and how people interact and experience the world. The overwhelming treatments add burdens to the patients and their families. In addition to physical suffering, cancer patients may incur psychiatric syndromes or experience worsening pre-existing conditions that are recognized as significant distress components [4,5]. The fear of death, and the anticipated losses, result in emotional distress, anxiety, and depression that may impact their quality of life [6,7]. There is evidence that untreated psychiatric comorbidities in patients with cancer significantly impact disability and quality of life, tending to worsen rapidly if not treated appropriately [8]. Enhancing the accountability of mental disorders in patients with cancer is essential for planning holistic care, improving the patient's quality of life, and

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tackling the resulting disability [9,10]. In addition, addressing these needs is vital throughout the continuum of cancer care [11].

It is important to focus on the psychological well-being of cancer patients. Despite the advances in treatment options and outcomes, psychological services and support have lagged behind for cancer patients [12]. It is noteworthy that psychological distress has a negative impact on the patients' quality of care and quality of life [13]. Studies have shown also that psychological distress among cancer patients has a negative impact on their physical health and treatment adherence [14,15]. Conversely, studies involving breast cancer women showed that consideration of the patients' psychological well-being improves treatment outcomes, including greater treatment adherence, better immune functioning, and improved overall survival [16,17].

With the rapid development of continuous care initiatives and a trend toward outpatient care of cancer treatments [6], technology has been introduced to cancer care to support patients' well-being and caregivers[18,19]. Technology has participated in improving cancer care by capturing patient-level data and using it in answering to support care standardization [19]. Patients can become more involved in their cancer with the help of health information technology such as telehealth, mobile health, and internet services [19]. For example, according to Eysenbach and colleagues, there may be four ways the internet could affect cancer patients: communication (e.g., e-mailing doctors), community (e.g., virtual support groups), health information on the web, and online shopping (e.g., ordering medications) [20,21]. In addition, cancer patients can benefit from using mobile health, as it can manage side effects, support drug adherence, provide cancer information, plan, follow up, and detect and diagnose cancer earlier with recognition of first signs. Patient education, information leaflets, patient diaries, and effective communication with healthcare professionals can help patients manage and report chemotherapy side effects [22]. Moreover, digital tools provide solutions to overcome logistical challenges contributing to cancer care disparities, with the potential to sublimate territorial and socio-economic barriers [23].

Many literature reviews explored the role of health technology in supporting care outcomes, communication, and quality of care [18,24,25]. However, little attention is given to the psychological well-being of cancer patients and to how technology can support their mental well-being. Therefore, it is essential to extend the existing knowledge base to understand better how digital health can support the mental health of patients with cancer. This systematic review aims to synthesize the qualitative and quantitative available literature on health information technology for mental health and psychological wellbeing support of cancer patients. We also try to identify the outcomes of technology use as well as enablers and barriers to implementation and use.

2. Materials and methods

We systematically reviewed the literature according to the Preferred Reporting Items for Systematic Reviews and meta-Analysis (PRISMA) guidelines. Our protocol was registered with the Open Science Framework on July 12th, 2022 [26].

2.1. Search strategy

Five distinct databases were systematically searched for relevant publications published between January 2011 and July 12th, 2022, representative of the biomedical, business, social sciences, humanities, education, science, and engineering and technology literature: ProQuest CENTRAL, PsycInfo, Scopus, PubMed, Web of Science, and IEEE Xplore. The dataset search was performed on July 12th, 2022, for all of them. The search strategy covered three broad areas: technology, care settings, and mental health. For each concept, we associated a range of keywords with enhancing the retrieval of relevant literature. The search terms used are summarized in Fig. 1. We combined these terms using the Boolean operators "AND/OR" to identify the relevant studies to the scope of the review through MeSH terms such as ("suicide" AND "patients" AND "cancer" AND "e-health"). These search words were identified by an initial review of the literature and then modified by feedback from content experts and the librarian. We exported the records retrieved to Endnote 20.1 (New Jersey, USA) for duplication removal and selection processes.

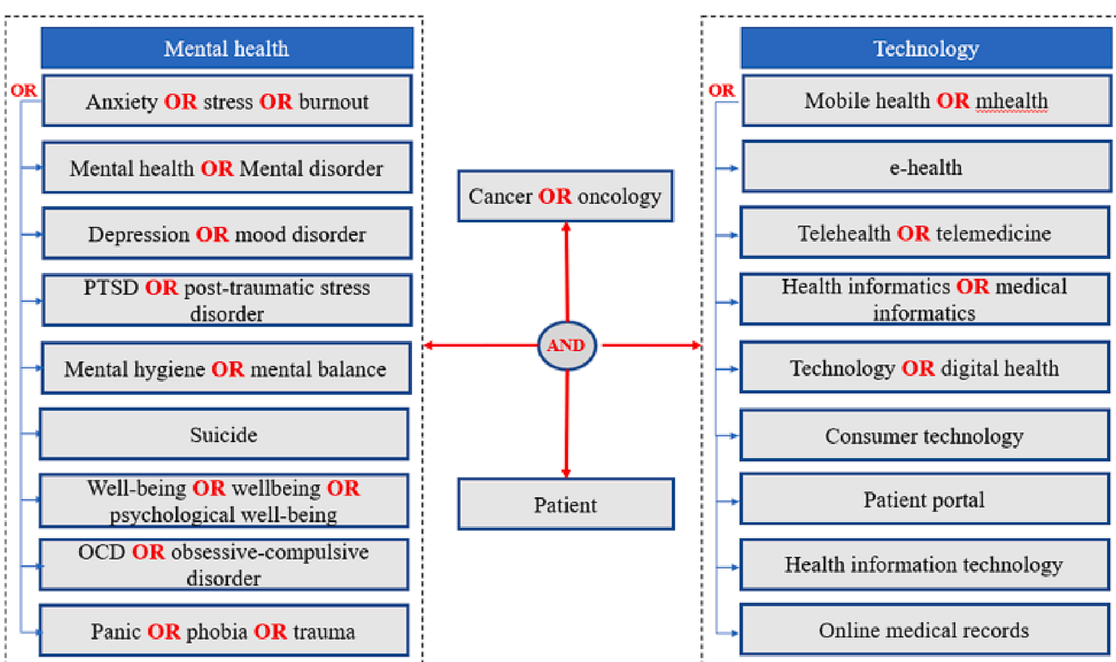

Fig. 1. Conceptual framework of the search terms used in the query of the studies for the review.

2.2. Eligibility criteria

Eligible papers are all study types presenting qualitative and/or quantitative empirical evidence, including surveys, interviews, experiments, and observational studies on the use of technology to support cancer patients' mental health. Only peer-reviewed publications written in English were included. We excluded opinion papers, editorials, and commentaries.

2.3. Study selection & quality assurance

Records were systematically extracted from 5 datasets, and duplicates were removed. First, we screened by reviewing the titles and abstracts (SE). Then, a full review of the text for the remaining articles was conducted. The PRISMA diagram illustrating the review flow is outlined in Fig. 2, and the number of articles in each step. A data abstraction form was used to record standardized information from each paper using Excel. All screening steps followed a consensus between the authors to minimize bias or discrepancy. For quality appraisal, we conducted a quality assessment of the papers following the Critical Appraisal Skills Program (CASP). The Critical Appraisal Skills Program (CASP) is a widely used and well-regarded quality appraisal tool for research studies [27]. The CASP tool provides a framework for critically appraising various types of research studies, including qualitative, quantitative, and mixed methods studies [27]. Appendices 1 and 2 summarize the quality assessment results run by the three authors.

3. Results

3.1. Data extraction and study characteristics

Our systematic review resulted in 5489 articles initially retrieved, with 192 duplicates: screening of the paper and full-text consultation ultimately resulted in 25 studies. Table 1 summarizes the objectives, years, study designs, outcomes measured, and technologies used in the selected papers.

3.2. Role of technology in mental health support among cancer patients

To report the results related to the technology's role in mental health support, we used the RYSS's Six-factor Model of Psychological well-being (SMPW). SMPW, a theory developed by Carol Ryff, determines the six main factors contributing to one's psychological well-being, contentment, and happiness: autonomy, environmental mastery, personal growth, positive relations with others, purpose in life, and self-acceptance [53]. Fig. 3 summarizes the conceptual model used to report the contributions of technology to mental health support among cancer patients.

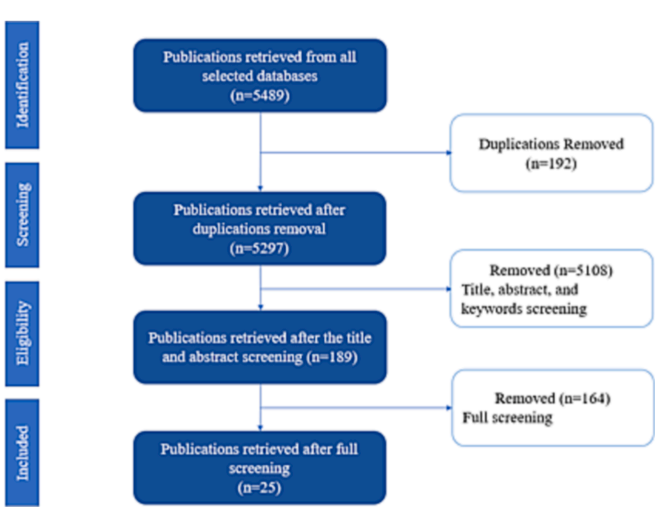

Fig. 2. PRISMA flow diagram of studies' selection.

As shown in Table 2 and Fig. 4, all the included studies (n = 25) reported that the technology used supported Personal Growth among cancer patients. Nineteenth supported their autonomy. Five supported environmental mastery, nine supported positive relationships, eight supported self-acceptance, and eventually, only two pursued patients' purpose and meaning in life. Some studies supported more than one theme, and others emphasized helping all of them.

Through the review results, we were able to validate the SMPW framework in capturing the role of technology in supporting cancer patients' psychological well-being.

The technologies used varied between mobile-based (mobile health, n = 12), web-based (web-based development program or intervention, n = 10; telehealth, n = 1; smart messaging, n = 1), and other digital devices (digital media device, n = 1).

To assess the impacts of digital devices on cancer patient's psychological well-being, the studies used different variables measured by different instruments and scales. Table 3 summarizes the outcome variables of the included studies with the measuring tools used. These metrics were utilized to report mental health outcomes in cancer patients.

By investigating the focus on sociodemographic factors in the studies included, we found that 24 % of the studies did not assess any demographic factors. Only 4 % of them assessed for the region (urban vs non-urban) and health literacy levels. 20 % focused on income levels, and 28 % accounted for race differences in their results. Fig. 5 shows the distribution in more detail.

3.2.1. Personal growth

Personal growth assesses one's openness to new experiences that can improve behavior and development [53]. A telehealth study showed that automated symptom monitoring coupled with pain and depression management interventions could improve the quality-adjusted life years and increase depression-free days among cancer patients [28]. Mindfulness and relaxation self-care apps could also enhance the quality of life by reducing anxiety and depression levels among cancer patients with depressive symptoms [33]. Another mobile-health system called CIMmH was used to provide psychological support based on meditation and stress management exercises for patients with esophageal cancer after undergoing esophagectomy [45]. The intervention showed the improved patient quality of life, stress, anxiety, and depression levels [45]. In addition to improving quality of life, STREAM, a web-based intervention designed to help cancer patients manage bodily, cognitive, and emotional stress, was effective in enhancing distress levels among newly diagnosed cancer patients [35].

3.2.2. Autonomy

Autonomy indicates the person's ability to regulate behavior independently of social pressures [53]. The selected studies showed that digital interventions significantly impact cancer patients' mental health. Web-based development programs such as MijnAVL provide personalized educational material for breast cancer survivors. Hope (Help To Overcome Problems Effectively) provides autonomy support for people living with cancer, showing improvement in self-efficacy and patient activation levels, and patient-empowerment among them [30,39]. Moreover, to improve autonomy, an app developed based on cognitive behavioral therapy used interactive features and videos that simulate interactions with therapists and patients with incurable Cancer [37]. Another app, HARUToday, promotes patients' self-management by resorting to psychoeducation, cognitive restructuring, and problem-solving training [38]. CALM, a mindfulness meditation app, was also used to improve patients' self-management to reduce anxiety and depression among patients with Myeloproliferative neoplasms [40]. Self-paced programs using mobile applications such as HeadSpacc are

Table 1
Synoptic summary of the data extracted from the systematic review.

Title	Year	Objective	Intervention	Technology	Study design	Outcomes measured
[28]	2014	Assessing the effectiveness of the telecare intervention INCPAD in support of the mental well-being of cancer patients	Telecare management trial, Indiana Cancer Pain and Depression (INCPAD), coupled with automated symptom monitoring for patients with cancer	Telehealth	Experiment and survey (N = 405 patients)	* Depression (Depression-free days (DFD), (PHQ-9)) * Anxiety and depression (Hopkins Symptom Checklist (HSCL-20)) * Quality of Life (Quality Adjusted Life Years (QALY) calculated out of depression, pain, mental health, and disability outcomes)
[29]	2014	Testing the effectiveness of WebChoice by comparing it to Internet-based patient-provider communication services (IPPC) for communication and health outcomes management	Web-based illness management support system WebChoice	Web-based development program	Experiment and survey (N = 176 patients)	* Symptom distress (Memorial Symptom Assessment Scale (MSAS-32)) * Anxiety (Hospital Anxiety and Depression Scale (HADS-A-7)) * Depression (Hospital Anxiety and Depression Scale (HADS-D-7)) * Self-efficacy (Cancer Behavioral Inventory (CBI-33))
[30]	2016	Evaluating the use, feasibility, and impact of the MijnAVL web portal among breast cancer survivors.	Interactive portal and video conference, MijnAVL	Web-based development program	Intervention and survey (N = 92 patients)	* Quality of Life (Short-Form 36-Item Health Survey (SF-36)) * Self-efficacy * Patient empowerment (patient activation measure (PAM))
[31]	2017	It tested the effectiveness of Mindful Self-Compassion interventions in supporting young and adult cancer survivors' management of psychosocial challenges in survivorship.	Web-based intervention adapted from the Making Friends with Yourself program	Web-based development program	Experiment and survey (N = 34 patients)	* Capacity of self-compassion (Self-Compassion Scale (SCS-26)) * Mindfulness (Mindful Attention Awareness Scale (MAAS-15)) * Anxiety, Depression, Social Isolation (PROMIS) * Relating to others, new possibilities, personal strength, spiritual change, appreciation of life (Posttraumatic Growth Inventory (PTGI-21))
[32]	2017	I am investigating the impact of 6-month and 12-month web-based interventions on emotional and social functioning, depression, and fatigue among cancer patients during recovery by providing psychosocial support and promoting positive lifestyle changes to improve quality of life.	The web-based computer-tailored Kanker Nazorg Wijzer (Cancer Aftercare Guide), KNW	Web-based development program	Interventions and survey (N = 231 patients)	* Emotional and social functioning (European Organization for the Research and Treatment of Cancer Quality of Life Questionnaire (EORTC QLQ-C30)) * Depression (HADS) * Fatigue (Checklist Individual Strength (CIS))
[33]	2018	Testing the acceptability and adherence to an app designed to help manage the mental needs of cancer patients with depressive symptoms	A mindfulness and relaxation self-care app	Mobile health	Mixed methods (N = 100 patients)	* Distress-Thermometer * Quality of life (FACT-G, Functional Assessment of Cancer Therapy-General) * Anxiety and depression (Hospital Anxiety Depression Scale (HADS)
[34]	2018	Testing the efficacy of a 2-week e-health stress inoculation training (SIT) intervention on emotion regulation and cancer-related well-being of elderly patients	Web-based intervention, eSIT	Web-based development program	Intervention and survey (N = 29 patients)	* Emotional coping (The Emotion Regulation Questionnaire ((ERQ-10), Cognitive reappraisal-6, Emotional suppression-4) * (Quality of Life) Emotional and social well-being (The Functional Assessment of Chronic Illness Therapy – Breast (FACT-B-29)) * Anxiety and relaxation (7-point visual analog scale)
[35]	2018	Evaluating the effectiveness of Web-based stress management (STREAM, Stress-Akti-Minden)	Web-based intervention, STREAM (Stress-Aktiv-Minden)	Web-based development program	Intervention and survey (N = 129 patients)	* Quality of Life (FACIT) * Distress (Distress Thermometer) * Anxiety and depression (HADS)
[36]	2019	Developing and testing a version of the Interaktor app adapted for patients undergoing pancreaticoduodenectomy.	Mobile health app (Interaktor)	Mobile health	Experiment and interview (N = 6 patients)	* Symptoms severity (fever, eating difficulties, nausea, vomiting, loose stool, constipation, pain, dizziness, fatigue, anxiety and worry, and problems with daily activities at home and outside the house, with frequency) * Risk assessment (risk-related alerts with different intensities)

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Table 1 (*continued*)

Title	Year	Objective	Intervention	Technology	Study design	Outcomes measured
[37]	2019	Testing the efficacy of a tailored cognitive-behavioral therapy (CBT) mobile application (app) to treat anxiety in patients with incurable cancer	Mobile application	Mobile health	Intervention and survey (N = 145 patients)	* Anxiety and depression, Quality of Life (HADS, Hamilton Anxiety Rating Scale (HAM-A), Patient Health Questionnaire-9, Functional Assessment of Cancer Therapy-General (FACT-G))
[38]	2019	It is developing and evaluating the effectiveness of an app-based cognitive behavioral therapy program for depression and anxiety in cancer patients.	A mobile-application-based cognitive behavioral therapy program (HARUToday)	Mobile health	Intervention and survey (N = 63 patients)	* Anxiety and depression (Beck Depression Inventory (BDI), State-Trait Anxiety Inventory (STAI)) * Quality of Life (Health-Related Quality of Life Scale (SF-36)) * Attitude (Dysfunctional Attitude Scale)
[39]	2020	Determining the ability of humorous digital media attention diversion to improve symptom domains of positive and negative mood during chemotherapy for patients with gynecologic cancers.	Digital media device with pre-loaded movies	Digital media device	Experiment and survey (N = 66 patients)	* Change in the mood (Positive/ Negative Affect Scale-Extended (PANAS-X) instrument) * Humor (Humor Styles Questionnaire (HSQ))
[40]	2020	Exploring the impact of mobile app use on depression, anxiety, and self-disturbance	A mindfulness meditation app, CALM	Mobile health	Experiment and survey (N = 80 patients)	* Depression, sleep disturbance, anxiety (Health Patient-Reported Outcomes Measurement Information System (NIH PROMIS)) * Mental health (Global Mental Health scale (GMH))
[41]	2020	Comparing the impacts of technology-delivered mindfulness-based intervention (MBI) programs; comparing the app-based intervention to a virtual classroom	Headspace is a self-paced program that provides guided mindfulness meditation instruction via a website or mobile application (iOS and Android) eMindful an online virtual classroom	Mobile health	Mixed methods (N = 103 patients)	* Quality of Life (Functional Assessment of Chronic Illness Therapy FACIT in Palliative care) * Distress (National Comprehensive Cancer Network Distress Thermometer) * Anxiety and depression (HADS) * Mindfulness (Five Facet Mindfulness Questionnaire (FFMQ-SF))
[42]	2020	Testing the usefulness of Stress Proffen, an app-based cognitive-behavioral stress management intervention for patients with cancer	Application for stress management, Stress Proffen	Mobile health	Experiment and survey (N = 25 patients)	* Stress (Perceived Stress Scale (PSS)) * Anxiety and Depression (HADS) * Health-Related Quality of Life - Short Form Health Survey (SF-36) * Fatigue (Self-Regulatory Fatigue)
[43]	2020	Exploring the usability and impact of a suite of mental health apps with phone coaching on psychosocial distress symptoms in patients recently diagnosed with breast cancer	Mental health coaching apps (IntelliCare)	Mobile health	Experiment and survey (N = 40 patients)	* Distress (Self-reported distress (Patient Health Questionnaire-4)) * Mood symptoms (Patient-Reported Outcomes Measurement Information System depression and anxiety scales)
[44]	2020	Evaluate the acceptability and preliminary efficacy of a novel app-based intervention with phone coaching in a sample of survivors of women's cancer.	Mental health coaching apps (iCanThrive)	Mobile health	Experiment and survey (N = 28 patients)	* Self-reported symptoms of depression (Center for Epidemiologic Studies Depression Scale). * Emotional self-efficacy * Sleep disruption
[45]	2020	Evaluating the feasibility, safety, and efficacy of a comprehensive intervention model using a mobile health system (CIMmH) in patients with esophageal cancer after esophagectomy.	Mobile health system (CIMmH)	Mobile health	Experiment and survey (N = 20 patients)	* Quality of Life (European Organization for Research and Treatment of Cancer-Quality of life Question-Core (EORTC-QLQ-C30, version 3.0), Esophageal Cancer Module (EORTC-QLQ-OES-18) questionnaires) * Depression (Chinese versions of Patient Health Questionnaire-9 (PHQ-9)) * Anxiety (General Anxiety Disorder-7 (GAD-7) * Stress (Stress Scale-10 (PSS-10))
[46]	2020	Evaluating the preliminary efficacy of a new web-based, psychoeducational distress self-management program, Caring Guidance™ after Breast Cancer Diagnosis, on newly diagnosed women's reported distress	Web-based intervention Caring Guidance	Web-based development program	Experiment and survey (N = 100 patients)	* Distress (Distress Thermometer (DT), Center for Epidemiologic Studies Depression Scale (CES-D), and the Impact of Events Scale (IES))
[47]	2020	Evaluating how women newly diagnosed with breast cancer used the unguided web-	Web-based psychoeducational intervention	Web-based development program	Interventions and survey (N = 54 patients)	* Distress level * Depressive symptoms

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Table 1 (*continued*)

Title	Year	Objective	Intervention	Technology	Study design	Outcomes measured
		based, psychoeducational distress self-management program Caring Guidance.				* Intrusive/avoidant thinking * Social constraints
[48]	2020	Examine an online-based cognitive therapy intervention's effectiveness and predict the factors that impact patients' adherence to such interventions.	Electronic Mindfulness-Based Cognitive Therapy (e-MBCT),	Web-based development program	Intervention and survey (N = 125 patients)	* Anxiety and depression (HADS-14) * Positive mental health (Mental Health Continuum Short Form (MHC-SF-14))
[49]	2020	Evaluating the effectiveness of smart messaging to improve medication adherence of patients with anxiety and depression.	Smart-messaging mindfulness-based cognitive therapy (MBCT) intervention	Smart messaging	Intervention and survey (N = 51 patients)	* Depression (PHQ-9) * Anxiety (GAD-7)
[50]	2021	Testing the feasibility of a digitally delivered self-management program for people with cancer to support mental well-being.	Digital self-management program, Help to Overcome Problems Effectively (HOPE)	Web-based development program	Experiment and survey (N = 61 patients)	* Mental well-being (Warwick Edinburgh Mental Well-being Scale (WEMWBS)) * Depression (9-item Patient Health Questionnaire (PHQ-9)) * Anxiety (Generalized Anxiety Disorder scale (GAD-7)) * Self-care confidence (Patient Activation Measure)
[51]	2021	Evaluating the effects and implementation of a mindfulness and relaxation app intervention for patients with cancer	A mindfulness and relaxation self-care app	Mobile health	Mixed methods (N = 100 patients)	* Mental and social health (Patient-Reported Outcomes Measurement Information System (PROMIS-29)) * Quality of Life (Functional Assessment of Cancer Therapy General (FACT-G)) * Distress (Distress Thermometer) * Mindfulness (Freiburg Mindfulness Inventory (FMI)) * Anxiety (Hospital Anxiety and Depression Scale (HADS-A-7)) * Depression (Hospital Anxiety and Depression Scale (HADS-D-7)) * Fear of progression (Fear of Progression Questionnaire-Short Form (FoP-Q-SF))
[52]	2021	Evaluating a supportive care initiative based on a mobile app for women with breast cancer to improve their quality of life	Educative mobile app training	Mobile health	Intervention and survey (N = 64 patients)	* Quality of Life (Functional Assessment of Cancer Therapy - Endocrine Symptoms Quality of Life Scale (FACT-ES QLS)) * Distress (National Comprehensive Cancer Network Distress Thermometer)

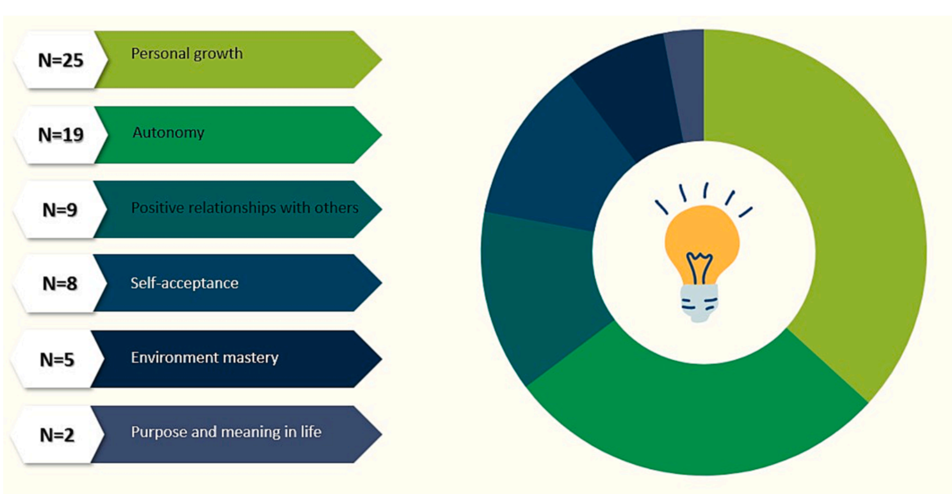

Fig. 3. Conceptual model of the technology's role in mental health support among cancer patients.

Table 2
Findings related to technology use and challenges of its adoption in cancer care for mental health support.

Title	The theme of well-being supported	Findings	Challenges of use and implementation	SEIPS factor challenged	Type of Cancer
[28]	Personal Growth	The telecare intervention improved the quality-adjusted life years and increased the depression-free days.	* Such technology requires a considerable time load and cost to implement in the beginning, but it decreases with time to only use cost	Organization Tools & Technology	breast cancer 29 %, lung 20 %, others 50 %
[29]	Personal Growth Autonomy Positive relationships with others	WebChoice has five components: - Diary where patients can take personal notes related to their situations - Communication where patients can share experiences with peers anonymously - Information where they can have access to reliable information and lifestyle suggestions from experts - Advice where they can manage their illnesses autonomously through suggested activities - Assessment where they can manage their psychological dimensions and symptoms. The system is designed to help cancer patients reduce their symptom distress, improve emotional well-being, and enhance self-efficacy. It also offers the chance for patients and doctors to communicate.	* The e-messages might disrupt existing workflows and increase workloads * A well-built technology requires more resources. It needs more people to be allocated to respond to users' concerns and communicate with them	People Tools & Technology Task Environment Organization	Breast cancer
[30]	Personal Growth Autonomy	The system includes educational material that is personalized to the patients' needs (physical, psychological, and social needs). Additionally, users receive a request by e-mail to complete patient-reported outcomes (PROs) about their quality of life at regular intervals with regular reminders.	* Patients' adherence to physical activity decreased over time. * It is hard to integrate such interactive systems into the care trajectory and the hospital information system	People Tools & Technology Environment Organization	Breast cancer survivors
[31]	Personal Growth Autonomy Environment Mastery Self-acceptance Positive relationships with others	The intervention helps the patients by introducing situations related to mindful self-compassion, paying attention to purposes, reactions to different situations, self-compassion in-depth, self-esteem support, finding one's voice, managing difficult emotions, and embracing life.	* Videoconferences have many environmental and technology-related challenges; internet connectivity, background noise, and some meditation music were hard to translate over videoconferences	Tools & Technology Environment	Young and adult cancer survivors
[32]	Personal Growth Autonomy Positive relationships with others	The intervention gave cancer patients a head start to psychological recovery after the end of cancer treatment. The intervention consists of eight modules and seven self-management training modules. The training modules cover returning to work, fatigue, anxiety and depression, social relationship and intimacy issues, physical activity, diet, and smoking cessation. The eighth module provides general information on the most common residual symptoms.	* The advice provided within the KNW on dealing with social relationships focuses less on support in coping with more complex social relationship structures, which impacts the patients' satisfaction and adherence to the intervention accounting for the different needs * Elderly patients are underrepresented in this study because they tend to be less prone to participate in digital and online interventions	People Tools & Technology	Cancer survivors
[33]	Personal Growth	The app showed high acceptance and adherence among patients, improving their quality of life, helping them manage their anxiety and depression	* Heterogeneous needs of patients	People	39 % breast cancer, 61 % other
[34]	Personal Growth Self-acceptance	The intervention is an e-health one based on SIT and was specifically designed and delivered online through a dedicated website. It is an online cognitive behavioral therapy designed to strengthen the patient's coping strategies to deal with stress. E-SIT has different phases; the conceptualization phase, where patients have face-to-face consultations with psychologists, and the skills acquisition phase, where they have interviews with other women who have gone through the same experiences and watch relaxing videos with guided meditation audios, and the application and follow-up phase, where they have interviews with patients undergoing chemotherapy with and without wigs together with relaxing guided meditations sessions. The improvement in patients' emotional well-being started three months after the intervention. E-SIT could help increase the levels of relaxation and anxiety among patients.	* Computer literacy may hinder the success of digital interventions * This digital intervention requires a long time of adherence to start giving good results, which may be costly and risk losing the patients' motivations * Designing eHealth interventions as part of regular care for breast cancer patients of all ages represents a challenge	People Tools & Technology	Breast cancer among elderly women
[35]	Personal Growth	This intervention is based on established stress management manuals incorporating cognitive behavioral and mindfulness-based stress reduction	* The intervention cannot effectively impact anxiety and depression among newly diagnosed cancer patients	People Tools & Technology	Newly diagnosed cancer patients

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Table 2 (*continued*)

Title	The theme of well-being supported	Findings	Challenges of use and implementation	SEIPS factor challenged	Type of Cancer
		techniques. STREAM includes modules that help manage stress (bodily, cognitive, and emotional), mindfulness and acceptance of thoughts and emotions, and activating resources such as quality of life, pleasure, social network, and communication. The app significantly improved the distress and quality of life among cancer patients.			
[36]	Personal Growth Autonomy Environment Mastery	The app has components that value symptoms management among cancer patients: (1) regular assessment of self-reported symptoms and problems, (2) a connection to a monitoring Web interface, (3) a risk assessment model for alerts on frequent or distressing symptoms, (4) continuous access to evidence-based self-care advice and links to relevant Web sites for more information, and (5) graphs for the patients to view a history of their symptom reporting. The patients felt reassured and cared for and received support for symptom management. Daily reporting of symptoms after pancreaticoduodenectomy enhances symptom management, self-care, and participation without burdening patients, indicating that mobile health can be used in clinical practice by patients with poor prognoses who experience severe symptoms.	* Some technologies don't assess the literacy levels of patients * Internet and server problems can add more burden to the user's experience and acceptance of the technology * Automating alerts for symptom management without healthcare providers' supervision can create confusion as some situations may be overrated and misguide the patient	Tools & Technology	Pancreatic Cancer
[37]	Personal Growth Autonomy	The app incorporated patient-centered, interactive, and personalized features, including videos simulating patient and therapist interactions, to teach skills for managing cancer-related anxiety. The CBT app helped improve cancer patients' anxiety, mood, and quality of life.	* Patients' adherence to the app-based interventions is still relatively low. * Mobile app interventions need more work on the design features and flexibility in how patients engage with the interventions' components to ensure more adherence	People Tools & Technology	Patients with incurable cancer
[38]	Personal Growth Autonomy	HARUToday comprises five zones: psychoeducation, behavioral activation, relaxation training, cognitive restructuring, and problem-solving.	* Some cancer patients do not prioritize mental health because of the challenges of the chronic and mortal diseases, which impacts the success of such technologies	People	Breast cancer 47.62 %, Other 52.38 %
[39]	Personal Growth	Using digital media attention diversion during chemotherapy significantly improved negative mood and fear. This was seen with both humorous and nonhumorous content. It is a low-cost, low-risk intervention that can be implemented easily and quickly in infusion centers.	* It is not possible to report the long-term effects of technology use due to the situation of cancer patients	People Task	Gynecologic
[40]	Personal Growth Autonomy	The app showed the potential to effectively help in self-management to reduce depression and anxiety symptoms in patients with cancer	* Interventions based on smartphones are lowly integrated into cancer care by providers in some hospitals	People Organization	Myeloproliferative neoplasm (MPN)
[41]	Personal Growth Autonomy	Patients preferred apps over virtual classrooms. The intervention showed improvement in anxiety, quality of life, and mindfulness. This intervention can be cost-saving for health care systems with limited resources to offer mental health services.	*Health care systems' scale impact the success of the technology-based interventions * Socio-economic status and geographic locations impact the adherence to technology	People Environment	Advanced Cancer
[42]	Personal Growth Autonomy Self-acceptance Positive relationships with others Purpose & meaning in life	The app-based intervention covers the following themes: Quality of Life and planning; Thoughts, feelings, and self-care; Mindfulness, rational thought-replacement, and guided imagery; Stress and coping; Social support, humor, and meditation; Anger management and conflict style awareness; Assertiveness and communication; and Health behaviors and setting goals. This study showed that mobile-based interventions could provide appreciated support for cancer survivors, should be easy to use, can significantly reduce stress, and improve emotional well-being.	* The app-based interventions need guidance from patients, which may add more workload on the patients, and providers by adding more tasks	People Task	Cancer survivors 40 % breast cancer, 60 % other
[43]	Personal Growth Autonomy Self-acceptance Positive relationships with others Purpose & meaning in life Environment Mastery	This app helps promote; awareness toward personal goals and values, identification of and communication with supportive individuals in one's life, challenging negative thinking patterns, coping with positive self-reinforcement, and mood management through physical activity. The app showed a significant decrease in general distress symptoms and symptoms of depression and anxiety postintervention.	* The numerous apps available online make it hard to allow for a better understanding of the effect of each app on psychosocial outcomes.	Tools & Technology	Newly diagnosed cancer patients

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Table 2 (*continued*)

Title	The theme of well-being supported	Findings	Challenges of use and implementation	SEIPS factor challenged	Type of Cancer
[44]	Personal Growth Autonomy Self-acceptance Environment Mastery	The app has eight modules that aim to support tenets of cognitive behavioral therapy (e.g., reducing worry and problem solving), acceptance-based therapies (e.g., mindfulness and emotional awareness), and positive psychology (e.g., fostering gratitude and savoring positive experiences). It helps patients challenge distorted thinking patterns. It also improves decision-making and problem-solving. In addition, it decreases worry and coping with related stress, increases relaxation stress, acknowledges things they are grateful for in daily life, identifies the emotions they are experiencing, the causal factors involved, and how their emotions are linked to their thoughts and thoughts behaviors. This intervention significantly decreases symptoms of depression and sleep disruption from baseline to postintervention	* Some individuals struggle to engage with the app, which impacts the findings related to the efficacy of the interventions	People Tools & Technology	Women with Cancer 46.42 % breast cancer, 32.14 % endometrial, 21.44 % Other
[45]	Personal Growth	The psychological support program of the app is based on meditation and stress management exercises adapted from mindfulness-based cancer recovery (MBCR) courses. Despite the app being designed for mental health support, the intervention should work on emphasizing the importance of mental health issues and educating patients to pay attention to their mental wellness after surgery more effectively.	* The intervention interface and content could be designed more interestingly and attractively than the actual one used	Tools & Technology	Esophageal Cancer
[46]	Personal Growth Autonomy Environment Mastery Self-acceptance Positive relationships with others	This new unguided, web-based psychoeducational program was developed to address the need for early and accessible self-management of cancer-related distress in newly diagnosed women to overcome institutional and patient barriers. It contains five modules; psychoeducation and cognitive-behavioral techniques, coping skills, problem-solving, communication strategies, and validation. This intervention can effectively reduce distress among women with breast cancer.	* Certain beliefs are hard to overcome to affect the patient's depressive symptoms	People	Breast cancer
[47]	Personal Growth Autonomy Positive relationships with others	The app supports patients by answering questions related to the following points: - problems, emotional reactions (fear, anger, and other emotional management strategies) - questions related to their self-acceptance (Self-concept Accepting support, the meaning of survivor, body image, and sexuality, people's reaction to one's preference - self-care strategies (Coping with Cancer, talking with people around you, personal control strategies) - moving-forward-related questions (Personal growth from this experience, cancer's impact on life, setting healthy goals). More access to sessions correlates with less distress. The key ingredient is not the amount of program use but each user's self-selected activity within the program (self-management).	* The videos embedded in the app did not help improve the patients' distress level. At the time of design, it was anticipated that women with more considerable distress at baseline would gravitate toward the program videos as an activity that took less focused attention. * The duration vs the frequency of use varies based on the patients' needs and cannot be used in a standardized way to evaluate the satisfaction with the app without accounting for the needs, constraints, and health situations	Tools & Technology	Breast cancer women
[48]	Personal Growth Autonomy Positive relationships with others	The intervention aimed to help cancer patients with psychoeducation about grief and cancer-related fatigue. Each session was spent on a specific theme: automatic pilot, communication, or self-care. Participants were provided information, audio files of guided meditation, and assignments around the session's music through a secure personal webpage.	* Adherence to digital interventions is not only determined by personal characteristics but also by environmental factors, support, and website characteristics * Patients need more support from mindfulness teachers to learn about used tools	People Environment Tools & Technology	Distressed cancer patients
[49]	Personal Growth Autonomy Self-acceptance	During MBCT, participants are trained to use mindfulness skills to address problems related to ruminative thinking patterns known to maintain anxiety and depression. The program focuses on using non-judgmental, present moment awareness to make purposeful choices about self-management of physical & emotional health.	* Patients find it hard to deal with telehealth and smart messaging * Interventions have to be customized to patients from individual cancer sites, stages, or treatment types	People Tools & Technology	Cancer survivors
[50]	Personal Growth Autonomy	The digital intervention program showed improved post-program scores on measures of positive mental well-being, depression, anxiety, and patient activation	* Patients' engagement drops through the progress of the sessions	People	41 % breast cancer, 19 % head or neck, 40 % other

(*continued on next page*)

Table 2 (*continued*)

Title	The theme of well-being supported	Findings	Challenges of use and implementation	SEIPS factor challenged	Type of Cancer
[51]	Personal Growth Autonomy Self-acceptance	The app offered three exercises: mindfulness meditation, guided imagery, and progressive muscle relaxation.	* The nurses often forgot about the app because it is not part of standard care * Not all patients possess smart devices to be part of such programs * The exercises provided are limited in content which may make the patients feel bored. The inclusion of exercises with different degrees of complexity and lengths of time was suggested	People Tools & Technology	Breast cancer 39 %, colon or ovarian 15 %, 40 % Others
[52]	Personal Growth Autonomy Positive relationships with others	The mobile app includes modules about the management of cancer side effects management, fatigue, mood changes, pain, sleep, relations issues, information seeking, and stress coping. In addition, it allows them to communicate with specialists, ask them questions, and remind them to record their symptoms and anxiety. The QoL of the treatment group after intervention increased, the distress level was lower compared to the control group, and the app was informative and valuable for patients.	* The app focused on improving the person's well-being, but there was a gap in social support. Thus, Social and Family Well Being were not impacted by the use of this app	Tools & Technology	Breast cancer women

*SEIPS: Systems Engineering Initiative for Patient Safety.

To report the results related to the technology's role in mental health support, we used the RYSS's Six-factor Model of Psychological well-being (SMPW). SMPW, a theory developed by Carol Ryff, determines the six main factors contributing to one's psychological well-being, contentment, and happiness: autonomy, environmental mastery, personal growth, positive relations with others, purpose in life, and self-acceptance [53]. **Figure 3** summarizes the conceptual model used to report the contributions of technology to mental health support among cancer patients.

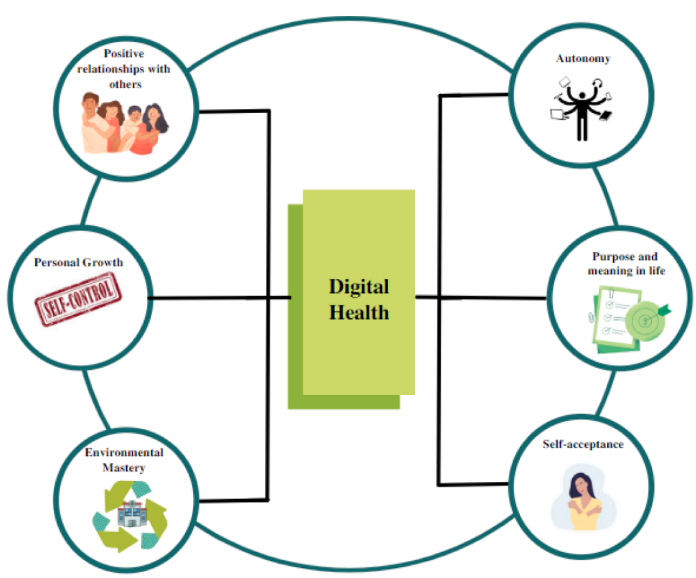

Fig. 4. Distribution of the technology's impact on the different themes of mental wellbeing.

Table 3
Tools used to assess the mental health and psychological well-being of cancer patients.

Anxiety and depression	• Hospital Anxiety Depression Scale (HADS) • Health Patient-Reported Outcomes Measurement Information System (NIH PROMIS) • Hopkins Symptom Checklist (HSCL-20) • 7-point visual analog scale
Anxiety	• Generalized Anxiety Disorder scale (GAD-7) • State-Trait Anxiety Inventory (STAI) • Hamilton Anxiety Rating Scale (HAM-A) • Hospital Anxiety and Depression Scale (HADS-A-7)
Attitude	• Dysfunctional Attitude Scale
Depression	• Patient Health Questionnaire (PHQ-9) • Center for Epidemiologic Studies Depression Scale for self-reported symptoms of depression • Depression-free days (DFD) • Beck Depression Inventory (BDI)
Distress	• Distress-Thermometer • National Comprehensive Cancer Network Distress Thermometer • Self-reported distress (Patient Health Questionnaire-4) • Hospital Anxiety and Depression Scale (HADS-D-7) • Memorial Symptom Assessment Scale (MSAS-32) • Center for Epidemiologic Studies Depression Scale (CES-D) • Impact of Events Scale (IES)
Coping	• Emotion Regulation Questionnaire (ERQ-10)
Fatigue	• Self-Regulatory Fatigue • Checklist Individual Strength (CIS)
Fear of progression	• Fear of Progression Questionnaire-Short Form (FoP-Q-SF)
Humor	• Humor Styles Questionnaire (HSQ)
Mental well-being/ Mental health	• Warwick Edinburgh Mental Well-being Scale (WEMWBS) • Global Mental Health scale (GMH) • Mental Health Continuum Short Form (MHC-SF-14) • Patient-Reported Outcomes Measurement Information System (PROMIS-29)
Mood	• Positive/Negative Affect Scale-Extended (PANAS-X)
Mindfulness	• Five Facet Mindfulness Questionnaire (FFMQ-SF) • Freiburg Mindfulness Inventory (FMI) • Mindful Attention Awareness Scale (MAAS-15)
Quality of Life	• Functional Assessment of Cancer Therapy-General (FACT-G) • Functional Assessment of Chronic Illness Therapy – Breast (FACT-B-29) • Functional Assessment of Chronic Illness Therapy - Palliative Care (FACIT-Pal) • Functional Assessment of Chronic Illness Therapy (FACIT) • Functional Assessment of Cancer Therapy - Endocrine Symptoms Quality of Life Scale (FACT-ES QLS) • Functional Assessment of Chronic Illness Therapy – Breast (FACT-B-29) • Short Form Health Survey (SF-36) • Quality Adjusted Life Years (QALY) calculated out of depression, pain, mental health, and disability outcomes • European Organization for Research and Treatment of Cancer-Quality of life Question-Core (EORTC-QLQ-C30, version 3.0 • Œsophage Cancer Module (EORTC-QLQ-OES-18) questionnaires
Self-care confidence, self-efficacy, the capacity for self-compassion, personal strength	• Patient Activation Measure (PAM) • Post-traumatic Growth Inventory (PTGI-21) • Self-Compassion Scale (SCS-26) • Cancer Behavioral Inventory (CBI-33)
Sleeping issues	• Sleep disruption scale • Health Patient-Reported Outcomes Measurement Information System (NIH PROMIS)
Stress	• Perceived Stress Scale (PSS)

also a cost-saving option for healthcare systems with limited resources to offer mental health services that engage patients in autonomously managing their well-being [41].

3.2.3. Positive relationship with others and environmental mastery

Environmental mastery refers to the effective use and management of environmental factors and activities [53], and positive relations with others refer to one's relationships that are built on empathy and affection [53]. Selected studies showed that it is possible to use technology to help cancer patients work on building better social life with positive relationships and master the environment around them. A web-based intervention called CaringGuidance was developed to effectively reduce distress among cancer patients by teaching them how to cope with their abilities to the different situations they may face and to build communication strategies that allow them to manage their interactions with their environment through the psychoeducational program and cognitive-behavioral techniques [46]. Mobile health's role was also significant as applications such as IteliCare, Interaktor, and iCanThrive significantly improved cancer patients' quality of life by empowering their environmental management skills [43–45]. For instance, by acknowledging things in daily life that they are grateful for, cancer patients that used iCanThrive were able to manage their well-being better, improve their sleep trends, and decrease symptoms of depression [44].

Furthermore, StressProffen, an app-based intervention for cancer survivors, could help them build a better social life and provide them with significant stress reduction and emotional well-being [42]. IntelliCare also helps a breast cancer patient communicate with the supportive individuals in her life to challenge negative thinking and build conflict awareness [43]. Another app-based intervention developed to support women with breast cancer showed improvement in distress level and quality of life [52]. The app allows breast cancer patients to communicate with specialists, ask them questions, and remind them to record their symptoms and anxiety [52]. It also helps them deal with their relations issues to manage their mood and stress [52], which is the same goal of electronic Mindfulness-Based Cognitive Therapy (e-MBCT) [48]. Finally, KNW (Kanker Nazorg Wijzer), a web-based cancer aftercare guide, showed success and effectiveness by supporting self-management among breast cancer patients by helping them solve their social relationship and intimacy issues [32].

3.2.4. Self-acceptance and purpose and meaning in life

Purpose in life indicates behavior improvement to empower patients on life's meaning and oneself value [53], and self-acceptance indicates one's positive attitude towards their personality [53]. In addition to the abovementioned benefits, digital health showed efficiency in supporting cancer patients' self-acceptance and goals. A smart-messaging mindfulness-based cognitive therapy (MBCT) intervention helps cancer survivors address issues related to their ruminative thinking patterns known to maintain anxiety and depression [49]. The program uses their awareness to help them make purposeful choices and manage their lives [49]. In addition, mobile health could support one's self-acceptance and purpose setting. StressProffen uses the patients' thoughts and feelings to

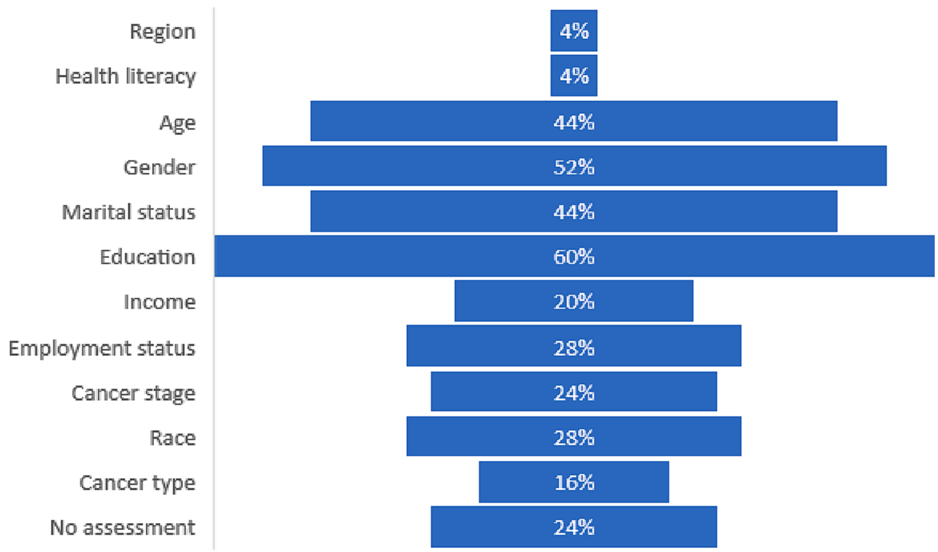

Fig. 5. The focus on sociodemographic factors in the studies selected.

build personalized strategies that help them manage their negative emotions and use them to construct a rational replacement way of thinking [42]. It also allows them to set goals and work on fulfilling them [42]. Acceptance-based therapies to support cognitive behavior by fostering gratitude and savoring positive experiences can help patients challenge their distorted thinking patterns and cope with their identity [44]. The intervention of mental health coaching (iCanThrive) could help patients decrease symptoms of depression and regulate their sleep disruption [44]. Using web-based interventions also allowed to promote cancer patients' self-acceptance. The Making Friends with Yourself (MFYP) program helps support one's self-esteem by introducing cancer survivors to situations related to self-compassion and helping them find their voices [31]. In addition, e-SIT, the emotional well-being support web-based intervention, helped breast cancer patients cope with their situations, thoughts, and emotions through educational content and interviews with other patients with the same disease or with a history of breast cancer [34].

3.3. Challenges of implementation and use in cancer settings

Despite the favorable impact of health information technologies in mental health support among cancer patients, it is noteworthy that several challenges still hinder its practical use and implementation. In this study, the challenges of technology use for cancer patients' mental health support are presented and discussed following the SEIPS model (Systems Engineering Initiative for Patient Safety). This theoretical model is rooted in person-centered systems engineering and human factors/ergonomics to improve healthcare processes [54]. Workers in healthcare institutions (including healthcare providers, biomedical engineers, and unit clerks) and patients are exposed to several health technologies and interact to deliver various tasks in daily life [54]. These tasks require a particular physical environment and organizational conditions [54]. Work system components (person, tasks, tools and technologies, physical environment, organizational conditions) interact and influence one another, producing different outcomes [54]. Fig. 6

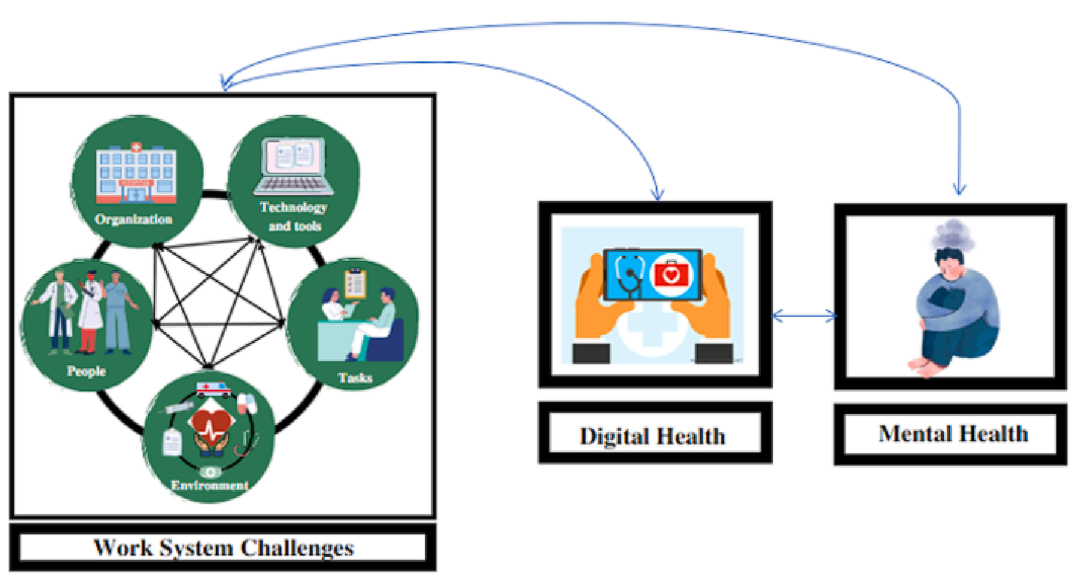

Fig. 6. The conceptual model of work system challenges of technology use for mental health support among cancer patients.

shows the conceptual framework used in this study to summarize the work system challenges impacting the use and implementation of digital health for cancer patients' mental health support.

3.3.1. Technology and tools

Our research identified some challenges related to the type of technology used. Some interventions cannot be effectively delivered because of internet and server issues or technical problems [31,36]. These barriers add burdensome to the users, impacting their user experience and the usability of tools [36]. In addition, some technology-based programs require a long term to start giving good results [34]. Such problems negatively impact the patients' adherence and make the programs costly in time and money [34]. Content-wise, some interventions lack personalization in the content delivered, which makes the patients unable to adhere to it. For example, the advice provided within the KNW on dealing with social relationships does not focus on supporting patients in dealing with more complex social relationship structures [32]. It does not account for the different needs of patients and, as a result, cannot satisfy all the users or comprehensively the single users [32].

Sometimes, the exercises are limited in content and may not completely satisfy expectations. Moreover, patients' uniqueness may not be captured by standardized and unflexible interfaces, so the design of the tools must assess the difference in health literacy levels between patients to deliver content understandable by all the participants [36]. Also, sophisticated reminders and alerts to help patients with symptom management or exercise adherence can yield confusion that could misguide them instead of assisting them [36]. Some technology features need more attention to the design and should give patients engagement flexibility to ensure good usability of the tool [37]. Finally, many technologies are online nowadays. It became hard for patients to decide which was best for their situations because they lacked clarity and guidance [43].

3.3.2. Organization and Environment

The technologies implemented are impacted by environmental and organizational challenges, such as the hospital's scale and geographic locations [41]. Implementing new technologies requires allocating resources that can respond to users' concerns and communicate with them, which exceeds some organizations' capacities [29]. Such a process can initially cost the organization money and time [28]. In addition, a technology adoption not framed for tackling specific tasks can disrupt the existing workflows and increase the workload among healthcare providers, nurses, and patients [29]. Furthermore, interactive systems are commonly hard to integrate into the care trajectory and in the information systems of the hospitals [30]. Patients have to use some technologies at home. Still, the issues to face may be beyond the simple use of the internet and intersect their level of education and, broadly, socio-economic determinants of health. The problems that they may face are overcome just by internet issues. Because of background noise and interruptions, patients cannot listen to the meditation music or the video conferences embedded in the interventions [31]. Finally, some hospital regulations and providers are not framed in a friendly way for mobile health adoption in cancer care, representing policy and normative barriers to its implementation [40].

3.3.3. People & tasks

Because of their critical health status, patients sometimes cannot adhere to long-term technology-based interventions [30,37]; in addition, patients have heterogeneous needs [33], challenging the overall adherence to digital health. The initial physical burden of cancer treatments may take over mental health needs, especially when patients are not appropriately advised and referred [38]. For example, newly diagnosed patients may experience a high burden of anxiety and depression that would benefit from stress management interventions [35]. Some patients' beliefs are hard to overcome and affect depressive symptoms to personal beliefs and non-adequate counseling [46]. Of note, the study of KNW's effectiveness had fewer elderly patients involved [32]. This was explained by the fact that this population tends to be less prone to participate in digital and online interventions and has less experience and education in digital technologies [32]. Designing electronic interventions suitable for all ages represents a significant challenge in making technology part of regular cancer care [34], as computer literacy can hinder the success of digital interventions [34]. Finally, the patients' socio-economic status and geographic locations impact the digital interventions' success [42].

4. Discussion

In our systematic review, we addressed a critical gap in digital health knowledge for cancer patients by comprehensively investigating its role and identifying gaps and potential barriers to implementation. We identified 25 studies focusing on health information technologies, using standardized frameworks to map the interventions based on the outcome pursued (Ryff model) and the themes of mental wellbeing addressed, and classifying the challenges of use-implementation based on the SEIPS model's components. To our knowledge, this is the first systematic review using standardized frameworks to map the field's literature comprehensively.

We identified that technology tools could enable cancer patients' personal growth, self-acceptance, and autonomy, emphasizing their purposes and importance [28,30,33,34,39,45]. It also allowed them to understand better the environments and the people around them [43–45,48,52]. It is essential to maintain self-acceptance high among cancer patients. Numerous studies have shown that cancer patients' self-esteem is a real risk factor for developing depressive disorders, often yielding to unhealthy habits [55]. Cancer patients with low self-acceptance will more likely struggle to cope with new strategies and routines [56]. Yet, high self-esteem can lead to a better sense of initiative and perseverance in facing difficulties [57,58]. Digital health interventions seem to have the potential to help patients support the way they perceive themselves. For example, StressProffen and iCanThrive, MFYP, among other interventions, succeeded in decreasing depression symptoms and sleep symptoms and improving quality of life by helping cancer patients find their voices heard, empowering them with self-compassion, building a purpose in life that drives them through their treatment and survivorship path [31,42,44].

Studies using pooled data showed that almost half of cancer patients reported high social and environmental problems [59]. It is hard for them to find support from families and friends and to deal with the challenges of their work and personal life [60–62]. Also, they have to face social isolation because of the social activity restrictions resulting from their disease [63]. Because cancer can seriously threaten social well-being, it is essential to forge patients' knowledge about the social and environmental factors that endanger their mental well-being. This can happen through interactive psychoeducation interventions, including digital health, to help patients communicate with other supportive individuals and challenge harmful environmental conflicts [43,53].

Most importantly, empowering patients' decision-making is essential in shaping their personal mental well-being growth and their quality of life improvement [64]. Shared decision-making maximizes their autonomy and accounts for their values and preferences [65]. To make decisions autonomously, patients need to be well-informed about the illnesses and the different options they have, together with the consequences of each. Health technology succeeds in offering these options through mobile health, web-based interventions, and virtual psychoeducation and coaching [30,35,39,41,45], which have all advantages of being accessible before and after the clinical visits, to refresh, educate, and persist in the information received. It is demonstrated that patients may retain limited information received during clinical visits, dampening the chances of fully informed, real shared decisions.

Despite the potential of technology to improve the mental well-being of cancer patients, we still need more work to improve dissemination, adoption, and engagement. Psychosocial and behavioral interventions have demonstrated efficacy for people with certain cancers [66,67], yet a clear gap exists between the treatment needs of survivors and the resources available in the community [68]. Improved dissemination and adoption of digital health technologies could play an important role in increasing the availability of mental health services for cancer survivors [69]. The limited availability of these services is a clear barrier to improving the mental health outcomes of cancer survivors [70,71]. Patients with access to mental health services frequently cite health care costs [66] as an additional barrier linked to distress [72]. Greater accessibility of digital health tools could offer cost-efficient alternatives for these patients.

As psychological and behavioral interventions have become more tailored to the needs of patients, digital health interventions must similarly embrace a person-centric approach that is both culturally sensitive and adaptable. There are well-documented racial and ethnic mental health disparities in which racial and ethnic minorities are less likely to receive adequate care [73]. These could be exacerbated without the thoughtful development of tools by both designers and clinicians. Digital health technologies can address logistical issues (i.e., patients unable to travel) or provide reminders via text message. However, tools must accommodate laptops and smartphones as many racial and ethnic minorities are more likely to use the latter to access the internet [74].

The digital tools offer avenues through which interventions can be further augmented. Smart devices are widely recognized as an effective tool for capturing patient-reported outcomes, yet they may also facilitate collecting psychophysiological processes such as heart rate variability [75,76]. This additional data may offer greater insight into patients' in-the-moment well-being, which could further inform treatment. For interventions related to exercise, an activity tracker, heart rate monitor, or smart watch that is paired with a digital health tool could accurately record adherence to the recommended exercises and tailor the intervention to offer options with different complexity and lengths of time.

Then, the complexity of the service availability may confuse patients and providers. Information about the various tools should be centralized. Technologies should be improved in design and content based on usability studies, but better navigation through options and opportunities should be framed. The tools' interfaces and content could be designed more interestingly and attractively to ensure patients understand what they are using, why they are using it, and how they will adhere to it.

Moreover, despite different technologies supporting the mental health of cancer patients, some support only specific themes of the Ryff's model. Customization will occur only if specific interventions match patients' needs. Finally, in addition to the technology suggested, it remains essential to ensure that the patients use it correctly. For that, more teaching support is needed from the nurses, healthcare providers, and caregivers.

Our study identified potential normative, regulatory, and policy barriers to implementation. We suggest that there cannot be any digital health impact without no political commitment and institutional will to embrace new technologies and frame functionally within workflows and harmonized, integrated, and person-centered pathways. Digital health interventions should not be another entry point to structural disparities. They should aim to improve access to mental health interventions, including for traditionally excluded or under-represented patients, to deliver the promise of mental health for all. In the context of the cancer care continuum, this should be a priority.

Our systematic review has clear limitations. Some studies have a limited sample size, which makes it hard to generalize the results to broader populations. Also, some have limited generalizability based on the quality assessment scores since some studies are feasibility and exploratory. More evidence is needed to drive general conclusions.

Most of the studies were based in high-income countries, so relevance, applicability, and outcomes should be interpreted through the lens of potentially high-function healthcare systems, albeit not free of radical disparities, with broader access to the internet, electricity, and/or smartphones. Second, we only surveyed the last decade of literature to discuss contemporary and relevant topics, given the rapid changes in technology capacities and approaches, and to capture the exponential increase in technology adoption for mental health support among cancer patients. Therefore, the findings from our synthesis may not reflect the experiences of earlier technological intervention in oncology settings. Moreover, the included studies targeted different types of cancer, making it hard to compare the samples. Finally, some populations (elderly patients, for example) are underrepresented. We could not specifically address minorities and other socio-economic disadvantaged populations. Still, the literature is pretty clear that implementing digital health insensitive to socio-economic impact will yield more disparities. So, the findings may not be generalizable to all patients and technology users. However, addressing digital determinants of health is the priority. Also, some of the studies were run during Covid-19, but more work should be done to compare the post and pre-covid to account for the impact of the pandemic and the lockdown on the technology use among cancer patients and their mental health. In this review, we did not aim to analyze it in detail.

Finally, by investigating the assessment of sociodemographic factors in the results of the studies, we found that many studies did not focus on minorities. Accounting for different levels of health literacy, race, and poverty levels, among other factors, is important in ensuring an effective technology design.

Our study has strengths. This is the first systematic review and a comprehensive approach based on standardized international frameworks. The study provides a broad landscape view of the evidence and ongoing challenges, identifying current potentials, limits, and unmet needs, therefore dissecting specific areas worthy of more research and technology work.

5. Conclusions

Our systematic literature review addressed a significant gap in cancer care literature. It focuses on digital health to support mental health among cancer patients. We found that different types of technologies have the potential to empower patients' personal growth, give them more autonomy and self-acceptance, and improve their environmental mastery and their social relationships, in addition to allowing them to set clear goals and find their lives' meanings. Future research needs to focus on improving the content and the design of the different tools suggested to maintain good mental health among cancer patients accounting for the difference in needs and use capabilities. Also, structural barriers should be tackled by formulating better, person-centered, and intersectoral health policies for digital health framed under the lens of equity. While some of the studies were run during Covid-19, more work should be done to compare the post and pre-covid to account for the impact of the pandemic and the lockdown on technology use among cancer patients and their mental health.

In conclusion, digital health has a transformative potential to deliver person-centric and individual, unique need-oriented health interventions to improve mental health, psychological well-being, and quality of life in patients with cancer.

6. Summary table

- Digital health has a transformative potential to deliver person-centric and individual, unique need-oriented health interventions to improve mental health, psychological well-being, and quality of life in patients with cancer.
- Health informatics have the potential to empower patients' personal growth, give them more autonomy and self-acceptance, and improve their environmental mastery and their social relationships, in addition to allowing them to set clear goals and find their lives' meanings.
- Future research needs to focus on improving the content and the design of the different tools suggested to maintain good mental health among cancer patients accounting for the difference in needs and use capabilities.
- Structural barriers should be tackled by formulating better, person-centered, and intersectoral health policies for digital health framed under the lens of equity.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1. Quality assessment of observational studies

	Selection	Comparability				Outcome		
	Representativeness of the included patients	Ascertainment of exposure	Demonstration that the outcome of interest was not present at the start of the study	Comparability of cohorts based on the design or analysis	Assessment of outcome	Follow-up time	Adequacy of follow-up	Final score
[40]	selected inclusion of users in a subgroup of the randomized patients that may lead to selection bias	structured interview *	yes, questionnaires are provided at baseline and post-4 weeks *	study controls for pre- and post- differences at baseline to endure no differences between the groups; multilevel factorial adjustment was applied; *	self-report	follow-up time is short, and the present analysis encompasses a 4-week interval	this analysis is performed on a subset of patients completing the questionnaires, so none is lost to the 4-week follow-up *	4*
[30]	selected patients, based on the voluntary will to participate, not selected by the investigators	a structured interview (MijnAVL) and medical records *	yes, questionnaires are provided at baseline and post-intervention *	adjustment for covariates and subgroup analyses are provided *	self-report (for the outcome) and medical records (for potential confounders)	not reported, no	not reported, no	3*
[33]	selected patients via online platforms may introduce selection bias skewed toward patients who are more confidential and comfortable with e-health applications	semi-structured interview to provide feedback on the tool but self-reported outcomes	baseline and post-baseline assessment *	model adjusted with a Cox proportional hazards regression *	self-report	The 10-week maximum time interval	longer follow-up data to be reported in an upcoming paper	2*
[42]	A selected group of volunteers from AMC and social media	structured interview *	Yes, outcome measured via questionnaire at baseline and post eight weeks *	The study does not control for any additional factors (baseline stress was also moderately correlated with attrition)	self-report	No, follow-up was eight weeks after the introductory session, but only 16 of 25 patients had viewed the majority (2/3) of app content by then	Three patients withdrew and were excluded from the final analysis	2*
[43]	Recruited breast cancer patients are receiving treatment from a Comprehensive Cancer Center. Somewhat representative of the average breast cancer patient in the community *	structured interview *	Yes, outcome measured via questionnaire at baseline and post seven weeks *	The study does not control for any additional factors	self-report	No, follow-up was 7 weeks after the initiation session	17 of 40 patients failed to complete the initiation visit	3*

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(*continued*)

	Selection	Comparability				Outcome		
	Representativeness of the included patients	Ascertainment of exposure	Demonstration that the outcome of interest was not present at the start of the study	Comparability of cohorts based on the design or analysis	Assessment of outcome	Follow-up time	Adequacy of follow-up	Final score
[44]	Patients selected from a larger community research cohort via mailing	structured interview *	Yes, outcome measured via questionnaire at baseline, 6 weeks, and 10 weeks *	The study does not control for any additional factors	self-report	No, follow-up was 4 weeks after the postintervention session (10 weeks from baseline)	Of the 28 who completed baseline, only 19 (68 %) completed the 4-week follow-up	2*
[45]	All patients meeting eligibility criteria were approached, minimizing the risk of bias*	no description	Yes, outcomes were measured via questionnaire at baseline, 1-month post-surgery, and 3 months post surgery*	The study does not control for any additional factors	record linkage and self-report mental health outcomes were self-reported, but other indicators of health were measured (weight, walking speed)*	Yes, the follow-up period was 3 months with measures repeated at month 1.*	4 of 20 patients were lost by the 3-month follow-up point. These patients were included in the 1-month results, and their attrition may have biased the 3-month results	4*
[36]	Participants were all from the same care group, which may lead to bias in the results	structured interview *	No questionnaires were provided at the start of the study	The study does not control for any additional factors	self-report	4 weeks is a short time for follow-up	1 patient dropped the study after 3 days at home	1*
[31]	no description of the derivation of the cohort	no description	Yes*	The study controls for sociodemographic factors*	self-report	Every week for 8 weeks	11 patients withdrew within 8 weeks	2*
[51]	Only a subgroup of the selected patients finalized the study	structured interview *	yes, questionnaires are provided at baseline and post 20 weeks *	study controls for pre- and post- differences at baseline, to endure no differences between the groups *	self-report	follow up time was set to daily for 20 weeks	No one withdrew from the study*	4*
[47]	no description of the derivation of the cohort	structured interview *	Yes, outcomes were compared to a baseline*	study controls for pre- and post- differences at baseline, to endure no differences between the groups *	self-report	follow up once per month for 3 months*	two participants withdrew after randomization	4*
[49]	patients selected from a group already attending the mindfulness-based cognitive therapy service may skew the selection toward more comfortable subjects for mental health applications and e-health	records (completion rate of the program) and interview (PHQ-9) *	analyses adjusted for baseline features *	a multilevel model that accounts for covariates *	independent analysis of an investigator not involved in patient care or administration of the interventions *	patients followed-up from start to the completion of the treatment *	no major dropouts *	6*
[50]	patients are enrolled in interventions according to their choice and not randomly	records (completion rate), self-reported outcomes (satisfaction) *	unclear	unclear	unclear	follow-up interval appropriate for the feasibility endpoint *	all patients enrolled were included in the primary analysis *	3*

Appendix 2. Quality assessment of randomized trials

	Randomization			Study Methodology			Were the study groups similar at the start of the randomized controlled trial?	Apart from the experimental intervention, did each study group receive the same level of care (that is, were they treated equally)?	Results			Relevance of the results		"YES" responses
	Did the study address a focused research question?	Was the assignment of participants to interventions randomized?	Were all participants who entered the study accounted for at its conclusion?	Were the participants 'blind' to the intervention they were given?	Were the investigators 'blind' to the intervention they were giving to participants?	Were the people assessing/analyzing outcome/s 'blinded'?			Were the effects of intervention reported comprehensively?******	Was the precision of the estimate of the intervention or treatment effect reported?	Do the benefits of the experimental intervention outweigh the harms and costs?	Can the results be applied to your local population/in your context?	Would the experimental intervention provide greater value to the people in your care than any of the existing interventions?	
[39]	Yes	yes	yes	yes	yes	yes	yes	yes	yes, based on structured scales (PANAS-X, BFI, HSQ)	DK results were reported as average changes with no confidence interval	yes	yes	DK	11
[50]	Yes	yes	yes	no	no	yes	yes	can't say	yes, structured scales	no, it is a feasibility study for a randomized trial. Unclear what will be the impact of the intervention	can't say, potentially, there is a special alert system in case of suicide risk	yes	can't say	7
[28]	Yes	yes	yes	DK	DK	Yes	DK	yes	yes, based on structured scales (SF-12, EQ-5D) and structured interviews	yes, results were reported with standard error	Yes	yes	DK	9
[46]	Yes	yes	yes	no	no	no	no, groups differed by income	yes	yes, based on structured scales (DT, CES-D, and IES)	Yes, results were reported with the standard deviations in intervals	DK	yes	DK	7
[38]	Yes	yes	yes	no	yes	yes	yes	yes	yes, based on structured scales (MSAS, HADS, CBI)	Yes, results were reported with confidence intervals	Yes	yes	DK	11
[52]	Yes	yes	yes	Yes	yes	yes	yes	yes	Yes, based on structured scales (such as FACT-ES QLS)	Yes results were reported with the standard deviations in intervals	Yes	yes	DK	12
[32]	Yes	yes	yes	yes	yes	yes	yes	yes	Yes based on structured scales (such as EORTC QLQ-C30, HADS, CIS)	Yes results were reported with the standard deviations in intervals	yes	yes	DK	12
[37]	Yes	yes	yes	yes	yes	yes	yes	yes	Yes based on structured scales (such as HADS, HAM-A)	Yes results were reported with the standard deviations in intervals	yes	yes	DK	12

(*continued on next page*)

(*continued*)

	Randomization			Study Methodology					Results			Relevance of the results		"YES" responses
	Did the study address a focused research question?	Was the assignment of participants to interventions randomized?	Were all participants who entered the study accounted for at its conclusion?	Were the participants 'blind' to the intervention they were given?	Were the investigators 'blind' to the intervention they were giving to participants?	Were the people assessing/analyzing outcome/s 'blinded'?	Were the study groups similar at the start of the randomized controlled trial?	Apart from the experimental intervention, did each study group receive the same level of care (that is, were they treated equally)?	Were the effects of intervention reported comprehensively?******	Was the precision of the estimate of the intervention or treatment effect reported?	Do the benefits of the experimental intervention outweigh the harms and costs?	Can the results be applied to your local population/ in your context?	Would the experimental intervention provide greater value to the people in your care than any of the existing interventions?	
[48]	Yes	yes	yes	no	no	no	yes	yes	yes, based on a standardized scale	DK	DK	yes	DK	7
[35]	Yes	yes	yes	yes	yes	Yes	Yes	Yes	Yes based on structured scales (such as FACT-ES, HADS)	Yes results were reported with confidence intervals	yes	yes	DK	12
[34]	Yes	yes	yes	no	no	no	yes	yes	yes, based on structured scales	yes	yes	yes	DK	9

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